CLL251: Heat Transfer for Chemical Engineers

Design of a Soil-Based Evaporative Milk Carrier

with Prospects for High-Strength Earthen Construction Materials

S.No.	Name	Entry Number
1	Pranay Kumar	2024CH70215
2	Nolin Kumawat	2024CH70990
3	Gowtham Naik	2024CH10178
4	Parthsarthi Singh	2024CH70625

Abstract

Milk preservation during transportation in rural and semi-urban regions remains a major challenge due to limited access to refrigeration and high energy costs. This work presents the design and thermal modeling of an evaporative-cooled annular milk storage container fabricated using porous soil materials.

The system employs passive evaporative cooling through a straw-water composite layer placed between concentric earthen walls. A transient thermal analysis of the milk and radial heat conduction through cylindrical layers is developed analytically.

The study predicts significant cooling below ambient conditions and demonstrates the potential of sustainable earthen construction materials for rural refrigeration applications.

1. Introduction

1.1 Background

Milk is highly perishable and requires controlled temperature conditions during storage and transportation. Traditional earthen pots have historically been used for evaporative cooling due to porous walls and natural evaporation effects.

1.2 Motivation

- Electricity-free cooling systems
- Sustainable and biodegradable materials
- Reduction of carbon emissions
- Low-cost rural cooling technologies

1.3 Objectives

- Design an evaporative-cooled annular milk carrier
- Develop analytical heat transfer equations
- Study transient milk temperature variation
- Analyze evaporative cooling performance

2. Problem Description and Geometry

The annular cylindrical container consists of:

1. Milk cavity
2. Inner soil wall
3. Straw-water composite layer
4. Outer soil wall
5. Ambient surroundings

Table 1: Geometric Dimensions

Symbol	**Description**	**Value**
r_1	Milk cavity radius	11 cm
r_2	Outer radius of inner soil wall	12 cm
r_3	Outer radius of straw-water layer	14 cm
r_4	Outer radius of outer soil wall	15 cm
L	Height	35 cm

3. Material Properties

3.1 Soil Properties

Property	Value	Unit
Thermal conductivity	0.8	W/mK
Density	1800	kg/m^3
Specific heat	880	J/kgK

3.2 Milk Properties

Property	Value	Unit
Density	1030	kg/m^3
Specific heat	3930	J/kgK
Thermal conductivity	0.55	W/mK

3.3 Prototype Configuration

The fabricated evaporative milk carrier prototype consists of concentric earthen walls separated by a straw–water composite layer used for passive evaporative cooling. The system is designed to maintain lower milk temperature during transportation without external electricity.

(a) Side view of fabricated earthen evaporative milk container

(b) Top cross-sectional view showing straw-water insulating layer

Figure 1: Experimental prototype of the annular evaporative milk carrier

4 Problem Description and Geometry

A milkman collects fresh milk from the cow at 5:00 AM when the ambient temperature is approximately 20 °C. The milk exits the cow at body temperature ($\approx$ 38.5 °C) and enters the container at $T_{m,0} \approx$ 35–37 °C after brief cooling during collection. The ambient temperature rises linearly to 45 °C by 12:00 PM. The container uses evaporative cooling through porous soil walls to keep the milk fresh.

4.1 Radial Geometry

The container is an annular cylinder with the following concentric layers (measured from the center outward):

Table 1: Radial geometry definitions

Symbol	Description	Range
$r \le r_1$	Milk (well-mixed cavity)	Inner region
$r_1 \le r \le r_2$	Inner soil wall (conduction)	Thickness $\delta_1 = r_2 - r_1$
$r_2 \le r \le r_3$	Straw + water composite (conduction)	Annular gap $\delta_g = r_3 - r_2$
$r_3 \le r \le r_4$	Outer soil wall (conduction)	Thickness $\delta_2 = r_4 - r_3$
$r > r_4$	Ambient air (convection + radiation)	External

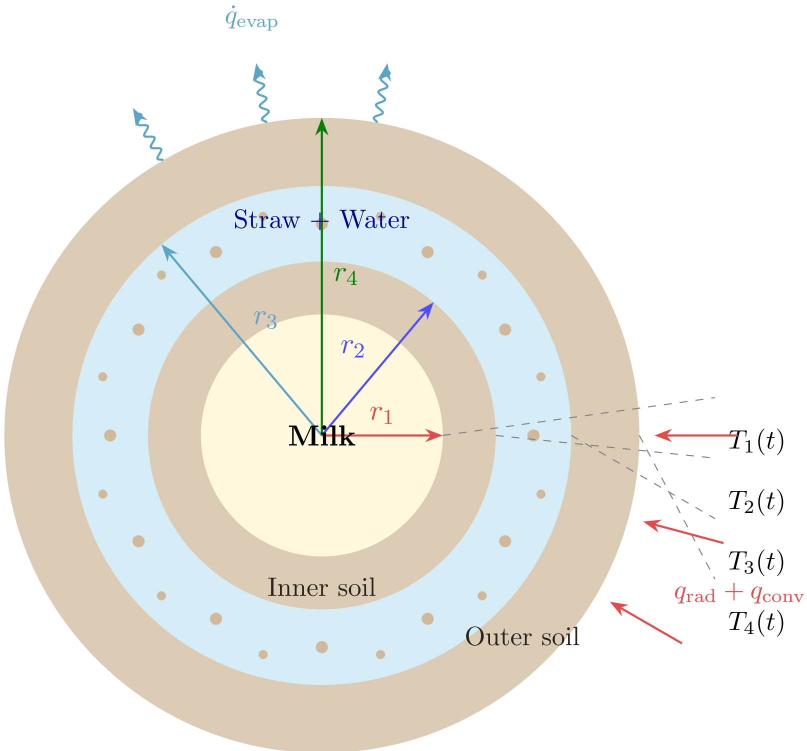

Figure 1: Cross-sectional view of the annular milk container showing radii r1 through r4, interface temperatures T1–T4, evaporative cooling, and ambient heat input.

5 Governing Equations

5.1 Heat Equation in Cylindrical Coordinates

For radial conduction in a cylindrical shell with no angular or axial variation:

$$\boxed{\frac{1}{r}\frac{\partial}{\partial r}\left(r\,k\,\frac{\partial T}{\partial r}\right) = \rho\,c_p\,\frac{\partial T}{\partial t}} \tag{1}$$

5.2 Quasi-Steady Approximation

The thermal diffusion time through the walls ($\sim \delta^2/\alpha$, order of seconds) is much shorter than the milk's thermal response time ($\sim mc_pR_{\text{total}}$, order of hours). Therefore, we adopt the **quasi-steady approximation**: the wall temperature profiles adjust instantaneously to the current boundary conditions, while only the milk temperature evolves transiently.

In each solid layer, the steady-state radial equation reduces to:

$$\frac{1}{r}\frac{d}{dr}\left(r\,\frac{dT}{dr}\right) = 0 \quad \Longrightarrow \quad T(r) = C_1 \ln r + C_2 \tag{2}$$

6 Time-Varying Ambient Temperature

The ambient temperature rises linearly from 5:00 AM to 12:00 PM:

$$\boxed{T_\infty(t) = T_{\infty,0} + \beta\,t} \tag{3}$$

where:

$$T_{\infty,0} = 20\,^\circ\text{C} \quad \text{(ambient at } t = 0\text{, i.e. 5:00 AM)} \tag{4}$$

$$\beta = \frac{T_{\infty,f} - T_{\infty,0}}{\Delta t} = \frac{45 - 20}{7} \approx 3.57\,\frac{^\circ\text{C}}{\text{hr}} \tag{5}$$

7 Thermal Resistance Network

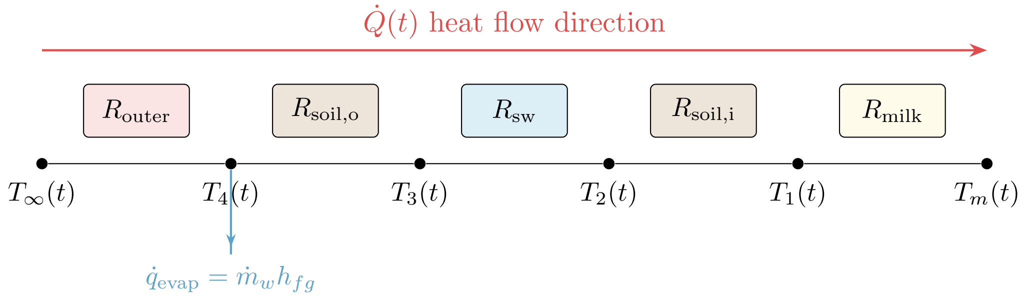

Figure 2: Thermal resistance circuit from ambient to milk with evaporative heat sink at the outer surface.

7.1 Resistance Expressions (Per Unit Length L)

$$R_{\text{outer}} = \frac{1}{2\pi\, r_4\, L\,(h_{\text{conv}} + h_{\text{rad}})} \qquad \text{(convection + radiation in parallel)} \tag{6}$$

$$R_{\text{soil,outer}} = \frac{\ln(r_4/r_3)}{2\pi\, k_{\text{soil}}\, L} \qquad \text{(conduction, outer soil wall)} \tag{7}$$

$$R_{\text{sw}} = \frac{\ln(r_3/r_2)}{2\pi\, k_{\text{eff}}\, L} \qquad \text{(conduction, straw+water composite)} \tag{8}$$

$$R_{\text{soil,inner}} = \frac{\ln(r_2/r_1)}{2\pi\, k_{\text{soil}}\, L} \qquad \text{(conduction, inner soil wall)} \tag{9}$$

$$R_{\text{milk}} = \frac{1}{2\pi\, r_1\, L\, h_{\text{milk}}} \qquad \text{(natural convection in milk)} \tag{10}$$

Total resistance:

$$\boxed{R_{\text{total}} = R_{\text{outer}} + R_{\text{soil,outer}} + R_{\text{sw}} + R_{\text{soil,inner}} + R_{\text{milk}}} \tag{11}$$

8 Temperature Distribution $T(r, t)$

8.1 Milk Temperature $T_m(t)$ — Transient ODE

Energy balance on the milk:

$$\boxed{m\, c_p \frac{dT_m}{dt} = \frac{T_\infty(t) - T_m(t)}{R_{\text{total}}} - \dot{m}_w\, h_{fg}} \tag{12}$$

Substituting $T_\infty(t) = T_{\infty,0} + \beta t$ and defining:

$$\tau = m\, c_p\, R_{\text{total}} \qquad \text{(thermal time constant)}, \qquad Q_{\text{evap}} = \dot{m}_w\, h_{fg} \tag{13}$$

the ODE becomes:

$$\tau \frac{dT_m}{dt} + T_m = \underbrace{(T_{\infty,0} + \beta t)}_{\text{rising ambient}} - \underbrace{Q_{\text{evap}}\, R_{\text{total}}}_{\text{evaporative depression}} \tag{14}$$

8.2 Analytical Solution

Using the integrating factor $e^{t/\tau}$:

$$\boxed{T_m(t) = \underbrace{\left[T_{\infty,0} + \beta t - Q_{\text{evap}}\, R_{\text{total}}\right]}_{T_{\text{eq}}(t)} - \beta\tau + \left[T_{m,0} - T_{\infty,0} + Q_{\text{evap}}\, R_{\text{total}} + \beta\tau\right] e^{-t/\tau}} \tag{15}$$

Physical interpretation of each term:

- $T_{\text{eq}}(t) = T_\infty(t) - Q_{\text{evap}}\, R_{\text{total}}$: instantaneous equilibrium temperature (ambient minus evaporative depression).
- $-\beta\tau$: thermal lag — the milk trails the ambient by $\beta\tau$ degrees because it cannot respond instantly.
- $e^{-t/\tau}$ term: transient decay of the initial temperature difference.

8.3 Interface Temperatures $T_1(t)$ through $T_4(t)$

The total heat transfer rate at any instant is:

$$\dot{Q}(t) = \frac{T_\infty(t) - T_m(t)}{R_{\text{total}}} - Q_{\text{evap}} \tag{16}$$

The interface temperatures are obtained by subtracting voltage drops across each resistance:

$$T_4(t) = T_\infty(t) - \dot{Q}(t) \cdot R_{\text{outer}} \tag{17}$$

$$T_3(t) = T_4(t) - \dot{Q}(t) \cdot R_{\text{soil,outer}} \tag{18}$$

$$T_2(t) = T_3(t) - \dot{Q}(t) \cdot R_{\text{sw}} \tag{19}$$

$$T_1(t) = T_2(t) - \dot{Q}(t) \cdot R_{\text{soil,inner}} \tag{20}$$

8.4 Radial Temperature Profile $T(r,t)$

Within each conduction layer, the logarithmic profile from Eq. (2) gives:

$$\text{Milk:} \quad T(r,t) = T_m(t) \qquad r \le r_1 \tag{21}$$

$$\text{Inner soil:} \quad T(r,t) = T_1(t) + \left[T_2(t) - T_1(t)\right] \frac{\ln(r/r_1)}{\ln(r_2/r_1)} \qquad r_1 \le r \le r_2 \tag{22}$$

$$\text{Straw+water:} \quad T(r,t) = T_2(t) + \left[T_3(t) - T_2(t)\right] \frac{\ln(r/r_2)}{\ln(r_3/r_2)} \qquad r_2 \le r \le r_3 \tag{23}$$

$$\text{Outer soil:} \quad T(r,t) = T_3(t) + \left[T_4(t) - T_3(t)\right] \frac{\ln(r/r_3)}{\ln(r_4/r_3)} \qquad r_3 \le r \le r_4 \tag{24}$$

Key feature: The temperature varies *logarithmically* with r (not linearly), which is the hallmark of cylindrical geometry. The time dependence enters through the interface temperatures $T_1(t), \ldots, T_4(t)$, which themselves depend on $T_m(t)$.

9 Heat Flux Distribution $q(r, t)$

The radial heat flux ($\mathrm{W/m^2}$, positive outward) is:

$$q(r,t) = -k\,\frac{\partial T}{\partial r} \tag{25}$$

Since $T(r,t) = A(t)\ln r + B(t)$ in each layer, $\partial T/\partial r = A(t)/r$.

$$\boxed{\begin{aligned}
&\text{Inner soil:} \quad q(r,t) = \frac{k_{\text{soil}}\left[T_1(t) - T_2(t)\right]}{r\,\ln(r_2/r_1)} && r_1 \le r \le r_2 \qquad (26)\\
&\text{Straw+water:} \quad q(r,t) = \frac{k_{\text{eff}}\left[T_2(t) - T_3(t)\right]}{r\,\ln(r_3/r_2)} && r_2 \le r \le r_3 \qquad (27)\\
&\text{Outer soil:} \quad q(r,t) = \frac{k_{\text{soil}}\left[T_3(t) - T_4(t)\right]}{r\,\ln(r_4/r_3)} && r_3 \le r \le r_4 \qquad (28)
\end{aligned}}$$

Important properties:
- $q(r,t) \propto 1/r$: heat flux is highest at the inner surface and decreases outward.
- Total heat rate $\dot{Q} = q(r,t) \cdot 2\pi r L = \text{const}$ through each layer (energy conservation).
- Sign: when $T_1 > T_2$, heat flows outward (positive q) — this is the desired cooling mode.

10 Evaporative Cooling Analysis

10.1 Energy Balance at the Outer

Surface At $r = r_4$, the energy balance is:

$$\underbrace{h_{\text{conv}}(T_\infty - T_4) + \varepsilon\sigma(T_{\text{sur}}^4 - T_4^4)}_{\text{heat in from ambient}} = \underbrace{q_{\text{cond}}\big|_{r=r_4^-}}_{\text{conduction into wall}} + \underbrace{\dot{m}_w'' h_{fg}}_{\text{evaporative loss}} \tag{29}$$

The evaporation depresses the surface temperature below ambient:

$$\boxed{T_4(t) \approx T_\infty(t) - \frac{\dot{m}_w\,h_{fg}}{2\pi r_4 L\,(h_{\text{conv}} + h_{\text{rad}})}} \tag{30}$$

In the limiting case of strong evaporation, $T_4 \to T_{\text{wet-bulb}}$.

10.2 Effect of Humidity

Evaporation rate depends on the vapor pressure deficit:

$$\dot{m}_w \propto (1 - \phi)\,h_m\left[p_{\text{sat}}(T_4) - \phi\,p_{\text{sat}}(T_\infty)\right] \tag{31}$$

where ϕ is the relative humidity and h_m is the mass transfer coefficient. As $\phi \to 1$, evaporation ceases and the cooling mechanism fails.

10.3 Conditions on Milk Temperature

Three regimes arise depending on the balance between ambient heating and evaporative cooling:

$$\frac{dT_m}{dt} = \frac{1}{mc_p}\left[\frac{T_\infty(t) - T_m}{R_{\text{total}}} - Q_{\text{evap}}\right] \tag{32}$$

Table 2: Operating regimes

Condition	Result	Physical meaning
$Q_{\text{evap}} > \dfrac{T_\infty(t) - T_m}{R_{\text{total}}}$	$\dfrac{dT_m}{dt} < 0$	Milk cools
$Q_{\text{evap}} = \dfrac{T_\infty(t) - T_m}{R_{\text{total}}}$	$\dfrac{dT_m}{dt} = 0$	Thermal equilibrium
$Q_{\text{evap}} < \dfrac{T_\infty(t) - T_m}{R_{\text{total}}}$	$\dfrac{dT_m}{dt} > 0$	Milk warms (slowly)

The equilibrium temperature (if it exists within the time window):

$$\boxed{T_{m,\text{eq}}(t) = T_\infty(t) - Q_{\text{evap}} \cdot R_{\text{total}}} \tag{33}$$

11 Graphs

11.1 Milk Temperature vs. Time (5 AM – 12 PM)

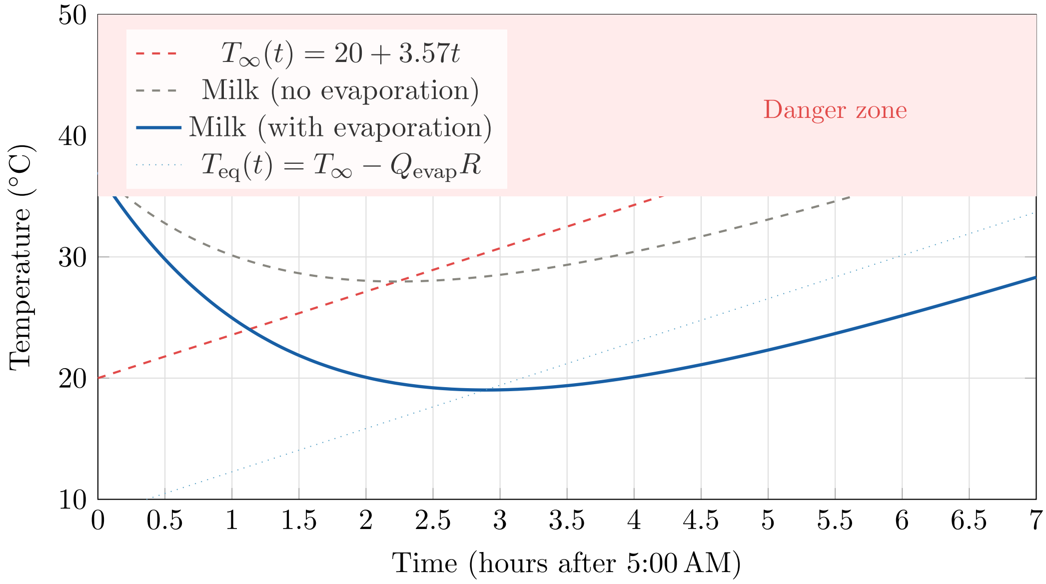

Figure 3: Temperature evolution of milk with and without evaporative cooling. Parameters: $m = 10\,\text{kg}$, $R_{\text{total}} = 0.15\,\text{K/W}$, $\dot{m}_w = 200\,\text{g/hr}$, $\phi = 40\%$, $T_{m,0} = 37\,°\text{C}$.

11.2 Radial Temperature Profile $T(r)$ at Selected Times

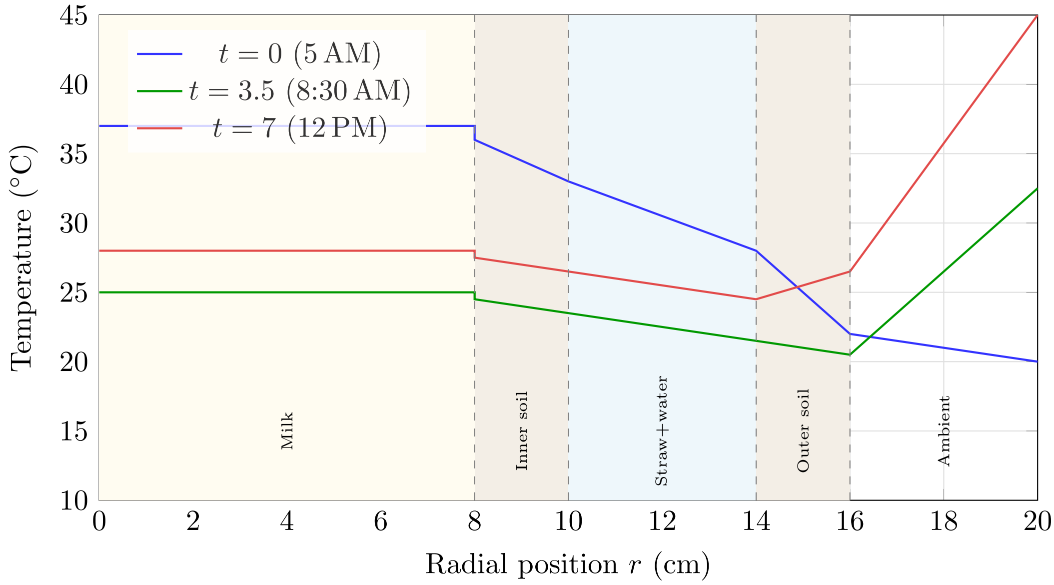

Figure 4: Radial temperature profile $T(r, t)$ at three time instants. Note the logarithmic shape in each conduction layer and the temperature inversion at $t = 7$ hr where the outer surface is cooler than ambient due to evaporation.

11.3 Heat Flux Profile $q(r)$

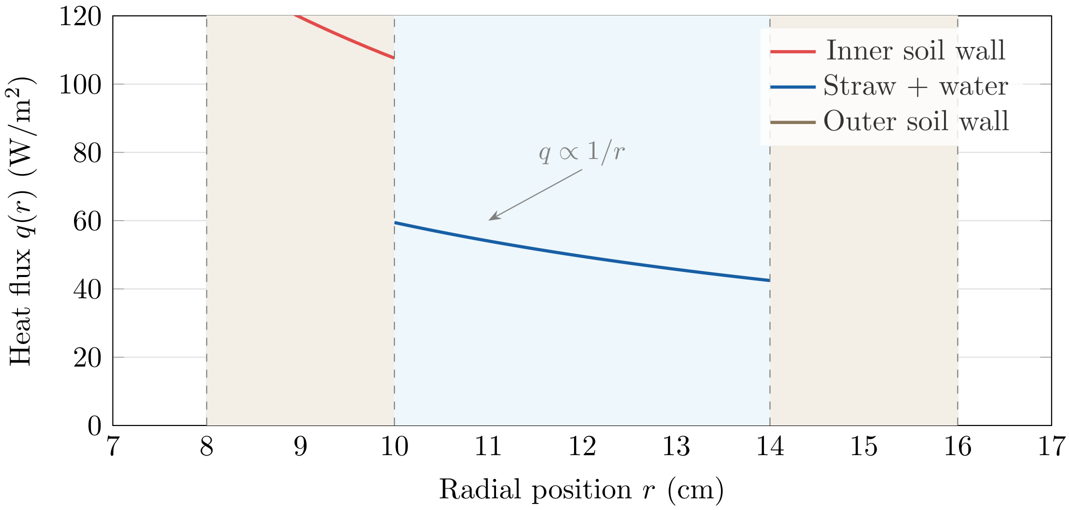

Figure 5: Radial heat flux profile $q(r)$ showing the $1/r$ dependence in each layer. Discontinuities at interfaces arise from different thermal conductivities.

Simulation Geometry and Meshing

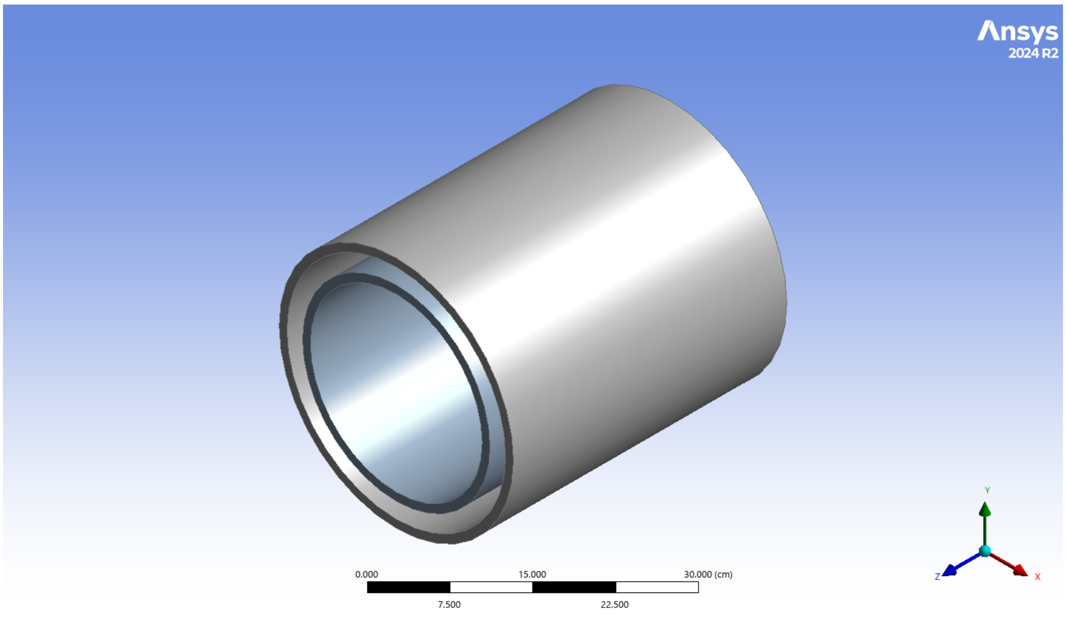

Figure 1: Simulation Geometry

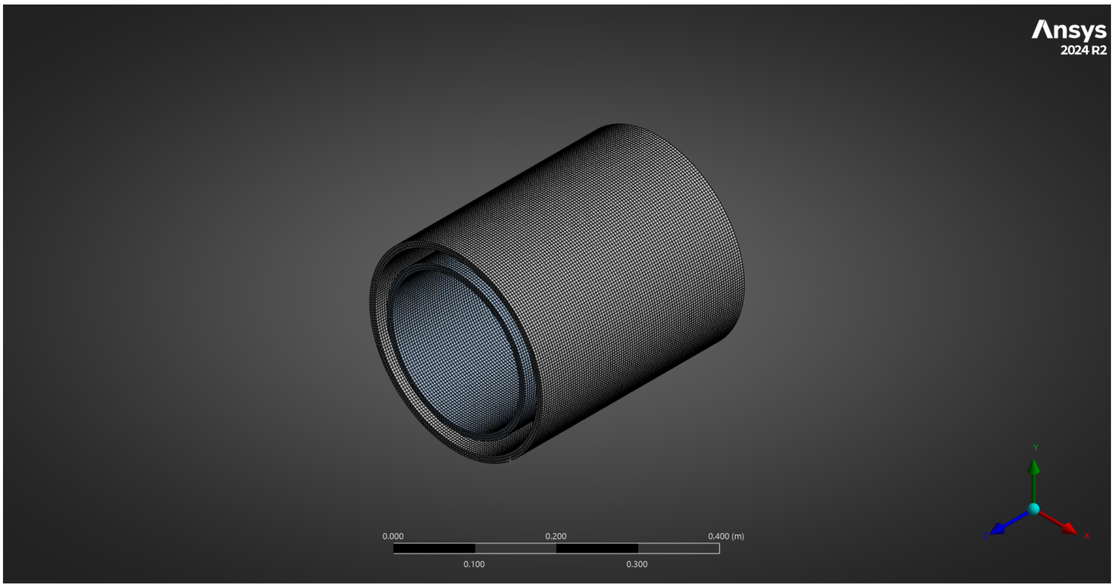

Figure 2: Simulation Meshing

12 Summary of All Formulas

Table 3: Complete formula summary

Quantity	Formula
Ambient temperature	$T_\infty(t) = T_{\infty,0} + \beta\, t$
Milk temperature	$T_m(t) = T_{\text{eq}}(t) - \beta\tau + \big[T_{m,0} - T_{\text{eq},0} + \beta\tau\big]e^{-t/\tau}$
Equilibrium temp.	$T_{\text{eq}}(t) = T_\infty(t) - Q_{\text{evap}} \cdot R_{\text{total}}$
Time constant	$\tau = m\, c_p\, R_{\text{total}}$
Heat rate	$\dot{Q}(t) = \big[T_\infty(t) - T_m(t)\big]/R_{\text{total}} - Q_{\text{evap}}$
$T(r,t)$ in walls	$T_{\text{in}}(t) + \big[T_{\text{out}}(t) - T_{\text{in}}(t)\big]\dfrac{\ln(r/r_{\text{in}})}{\ln(r_{\text{out}}/r_{\text{in}})}$
$q(r,t)$ in walls	$\dfrac{k\big[T_{\text{in}}(t) - T_{\text{out}}(t)\big]}{r\,\ln(r_{\text{out}}/r_{\text{in}})}$

13 Typical Parameter Values

Table 4: Recommended parameter values for design calculations

Parameter	Symbol	Value	Unit
Milk specific heat	c_p	3930	J/(kg·K)
Milk density	ρ_m	1030	kg/m^3
Milk initial temp.	$T_{m,0}$	37	°C
Soil thermal cond.	k_{soil}	0.5–1.5	W/(m·K)
Straw+water eff. cond.	k_{eff}	0.3–0.6	W/(m·K)
Latent heat of water	h_{fg}	2260	kJ/kg
Outer convection coeff.	h_{conv}	5–15	W/(m^2·K)
Radiation coeff.	h_{rad}	4–8	W/(m^2·K)
Milk nat. conv. coeff.	h_{milk}	50–200	W/(m^2·K)
Water evap. rate	$\dot{m}_w$	100–300	g/hr

14 Complete List of Assumptions

Every mathematical model involves simplifications o f r eality. B elow i s a comprehensive, categorised list of *every* assumption embedded in the formulation presented in this report. For each assumption we state **what** was assumed, **why** it was made, and **what would change** if the assumption were relaxed.

14.1 Geometric Assumptions

G1. Infinite cylinder (no end effects). We model heat transfer only in the radial direction (r), ignoring axial (z) and circumferential (θ) variations. This is valid when $L/r_4 \gg 1$ (tall container relative to its radius). In practice the top and bottom surfaces of the container also exchange heat with the surroundings; a correction factor or separate 1-D model for the lid/base can be added.

$$\frac{\partial T}{\partial \theta} = 0, \qquad \frac{\partial T}{\partial z} = 0 \quad \Longrightarrow \quad T = T(r,t) \text{ only} \tag{G1}$$

G2. Perfect concentricity. All cylindrical layers are assumed to be perfectly concentric with uniform wall thicknesses. In a handmade clay container, eccentricity would create thinner/thicker wall regions, producing angular temperature variation and preferential heat flow paths.

G3. Constant radii (no thermal expansion). The radii r_1, r_2, r_3, r_4 are treated as fixed constants. Thermal expansion of the soil walls and milk is negligible at the temperature differences involved ($\Delta T \sim 20\,°\mathrm{C}$, $\alpha_{\mathrm{linear,clay}} \sim 5 \times 10^{-6}\,\mathrm{K}^{-1}$).

G4. Smooth interfaces. The interfaces between layers (soil–water, soil–milk) are assumed to be geometrically smooth. In reality, the soil surface is rough and porous, creating an irregular contact with the adjacent fluid. This is absorbed into the effective heat transfer coefficients.

G5. Uniform cross-section along the height. The container is modeled as having a constant cross-section from bottom to top (a true cylinder). If the container is tapered, bulged, or has a narrower neck, the cross-sectional areas and radii would vary with z, requiring integration along the height.

14.2 Thermal Property Assumptions

P1. Constant thermal conductivity (homogeneous, isotropic materials). k_{soil} and k_{eff} are treated as constants, independent of temperature and position. In reality:

- k_{soil} depends on moisture content (wet clay $\approx 1.5\,\mathrm{W/mK}$ vs. dry clay $\approx 0.5\,\mathrm{W/mK}$). As water seeps through the wall, k varies with r.
- The straw+water layer is anisotropic — k depends on whether heat flows along or across the straws.

Relaxing this requires solving the variable-conductivity equation $\frac{1}{r}\frac{d}{dr}\big(r\,k(r)\frac{dT}{dr}\big) = 0$, which does not yield a simple logarithmic profile.

P2. Constant specific heat and density of milk. $c_p = 3930\,\mathrm{J/(kg{\cdot}K)}$ and $\rho_m = 1030\,\mathrm{kg/m^3}$ are taken as constants. Milk properties vary slightly with temperature and fat content (c_p changes by $< 3\%$ over 20–40 °C), so this is an excellent approximation.

P3. Straw+water layer treated as an effective homogeneous medium. The composite of straws (hollow tubes of organic material) saturated with water is replaced by a single material with effective thermal conductivity k_{eff}. This assumes:

- Straws and water are in local thermal equilibrium (no temperature difference between straw material and adjacent water at the same r).

- The effective medium is isotropic (same k_{eff} in all directions).

More accurate models (e.g. Maxwell, Bruggeman, or parallel/series composites) can be used to estimate k_{eff} from the porosity ε and individual conductivities:

$$k_{\text{eff}} \approx \varepsilon\, k_{\text{water}} + (1-\varepsilon)\, k_{\text{straw}} \quad \text{(parallel model, upper bound)} \tag{P3}$$

P4. Inner and outer soil walls have the same thermal conductivity. Both walls are assumed to be made of the same soil/clay material. If different clays or different moisture levels exist, separate k values should be used.

P5. Latent heat of vaporisation h_{fg} is constant. $h_{fg} = 2260\,\text{kJ/kg}$ is used at $100\,^\circ\text{C}$. At the actual evaporation temperature (~ 20–$30\,^\circ\text{C}$), $h_{fg} \approx 2440\,\text{kJ/kg}$. This introduces a $\sim 8\%$ underestimate of the evaporative cooling power. Using the temperature-corrected value is straightforward.

14.3 Heat Transfer Mode Assumptions

H1. Pure conduction in all solid/composite layers. Heat transfer through the inner soil wall, straw+water layer, and outer soil wall is assumed to be purely by conduction. Natural convection within the water-filled annular gap is neglected. This is justified when:

- The gap is narrow (Rayleigh number $\text{Ra}_\delta < 1000$ for the annular gap).
- The straws act as flow suppressors, damping buoyancy-driven circulation.

If the gap is wide and straws are sparse, free convection in the water layer would enhance heat transfer beyond what k_{eff} predicts.

H2. Radiation and natural convection in parallel at the outer surface. At Interface 1 (outer surface $r = r_4$), both radiation and convection act simultaneously between the surface and the surroundings, sharing the same temperature difference $(T_\infty - T_4)$. They are therefore parallel resistances:

$$h_{\text{total,outer}} = h_{\text{conv}} + h_{\text{rad}} \tag{H2}$$

This is standard for external surfaces. The linearised radiation coefficient is:

$$h_{\text{rad}} = \varepsilon\,\sigma\,(T_4^2 + T_\infty^2)(T_4 + T_\infty) \approx \varepsilon\,\sigma\,4\,T_{\text{avg}}^3 \tag{H2b}$$

where ε is the emissivity of the soil surface (≈ 0.9 for clay).

H3. Linearised radiation. The $\sigma(T_{\text{sur}}^4 - T_4^4)$ radiation term is linearised into $h_{\text{rad}}(T_\infty - T_4)$. This is valid when $|T_\infty - T_4| \ll T_{\text{avg}}$ (in Kelvin), which is well satisfied here (temperature differences ~ 10–$20\,\text{K}$ on a base of $\sim 300\,\text{K}$).

H4. Natural convection inside the milk cavity. The milk is treated as a well-mixed fluid with uniform temperature $T_m(t)$, exchanging heat with the inner soil wall via a natural convection coefficient h_{milk}. This is a lumped treatment of the milk. A more detailed model would solve the Navier-Stokes equations inside the milk cavity to find the actual velocity and temperature fields, but for engineering design this lumped approach is standard.

H5. Constant heat transfer coefficients. h_{conv}, h_{rad}, and h_{milk} are assumed constant throughout the process. In reality:

- h_{conv} depends on ΔT: for natural convection, $h \propto (\Delta T)^{1/4}$ (laminar) or $(\Delta T)^{1/3}$ (turbulent).

- h_{rad} depends on $(T_4^2+T_\infty^2)(T_4+T_\infty)$, which changes as temperatures evolve.
- h_{milk} depends on the Rayleigh number inside the milk, which changes as the driving ΔT evolves.

For a more accurate model, these coefficients should be updated at each time step using appropriate correlations.

H6. No forced convection. The outer surface convection is assumed to be purely natural (buoyancy-driven). During transport on a bicycle or cart, forced convection would significantly increase h_{conv} (possibly from 5–10 to 20–50 W/m^{2}K), which would: (a) increase heat gain from ambient when $T_\infty > T_4$, but also (b) increase evaporation rate by removing the boundary layer of humid air.

H7. No contact resistance at interfaces. Perfect thermal contact is assumed between all layers (soil–water, soil–milk). In reality, air gaps, surface roughness, or poor wetting could introduce additional thermal resistance at each interface:

$$R_{\text{contact}} = \frac{1}{h_c \cdot A} \tag{H7}$$

For wetted clay surfaces, contact resistance is typically negligible.

14.4 Boundary Condition Assumptions

B1. Linearly rising ambient temperature.

$$T_\infty(t) = T_{\infty,0} + \beta\, t, \qquad \beta = \frac{45-20}{7} \approx 3.57\ {}^\circ\text{C/hr} \tag{B1}$$

In reality, the diurnal temperature cycle is better modeled by a sinusoidal function:

$$T_\infty(t) = T_{\text{mean}} + \Delta T\ \sin\left(\frac{2\pi}{24}(t - t_{\text{peak}} + 6)\right) \tag{B1b}$$

However, the linear ramp is a good approximation during the 5 AM–12 PM morning window when temperature rises monotonically. The linear model is conservative (predicts slightly faster temperature rise than reality in some periods).

B2. Uniform ambient temperature around the container. The same $T_\infty(t)$ is applied to the entire outer surface. In practice, the side facing the sun would be hotter than the shaded side, creating circumferential variation. For conservative design, using the maximum expected ambient temperature is appropriate.

B3. No direct solar radiation absorbed. The model treats the outer surface heat gain as convection + radiation from surroundings at temperature T_∞. Direct solar irradiation G_{solar} is not separately modeled. If the container is in direct sunlight:

$$q_{\text{solar,absorbed}} = \alpha_s\, G_{\text{solar}} \quad (\alpha_s \approx 0.6\text{–}0.8 \text{ for clay, } G_{\text{solar}} \approx 800\text{–}1000\ \text{W/m}^2) \tag{B3}$$

This can be incorporated as an additional heat source at the outer surface or by using the sol-air temperature $T_{\text{sol-air}} = T_\infty + \alpha_s G/h_{\text{outer}}$ instead of T_∞.

B4. Top and bottom surfaces are adiabatic. No heat transfer through the lid or base is considered. This follows from Assumption G1 (no axial variation). In reality, the lid (especially if not soil-based) can be a significant heat entry path, and the base resting on a surface conducts heat depending on the surface material and temperature.

B5. Initial milk temperature is uniform. At $t = 0$, the milk is at a uniform temperature $T_{m,0} = 37\,°\mathrm{C}$ throughout. This is reasonable because the milk is stirred/poured into the container, and natural convection quickly homogenises the temperature within the liquid.

B6. Initial wall temperatures match a steady profile. The wall temperature profiles at $t = 0$ are assumed to be the steady-state profiles corresponding to $T_m(0)$ and $T_\infty(0)$. Since the walls reach thermal equilibrium in seconds (Assumption QS), this is self-consistent.

14.5 Temporal / Dynamic Assumptions

T1. Quasi-steady wall conduction. The thermal diffusion time through each wall is much shorter than the system time constant:

$$t_{\text{diff,wall}} = \frac{\delta^2}{\alpha_{\text{soil}}} \sim \frac{(0.02)^2}{5 \times 10^{-7}} \approx 800 \text{ s} \approx 13 \text{ min} \tag{T1a}$$

$$\tau_{\text{system}} = m\, c_p\, R_{\text{total}} \sim 10 \times 3930 \times 0.15/3600 \approx 1.6 \text{ hr} \tag{T1b}$$

Since $t_{\text{diff}} \ll \tau_{\text{system}}$, the walls "track" the boundary conditions quasi-statically. For very thick walls ($\delta > 5\,\mathrm{cm}$) or very light containers, this assumption may weaken.

T2. Lumped capacitance for the milk (Bi < 0.1). The milk is treated as having a single, uniform temperature $T_m(t)$ at each instant. This requires the internal resistance of the milk to be small compared to the convective resistance at its surface:

$$\text{Bi}_{\text{milk}} = \frac{h_{\text{milk}} \cdot r_1}{k_{\text{milk}}} \ll 1 \tag{T2}$$

With $h_{\text{milk}} \sim 80\,\mathrm{W/m^2K}$, $r_1 \sim 0.08\,\mathrm{m}$, $k_{\text{milk}} \approx 0.55\,\mathrm{W/mK}$, we get Bi ≈ 11.6, which is *not* small. This means the milk actually has internal temperature gradients. However, natural convection mixing inside the liquid tends to homogenise the temperature, making the lumped model a reasonable engineering approximation. A more rigorous approach would couple the internal convection problem.

T3. Constant evaporation rate $\dot{m}_w$. The mass flow rate of water evaporating from the outer surface is assumed constant over the 7-hour period. In reality:

- As the day gets hotter and drier, evaporation rate increases.
- As the water reservoir (straw+water layer) depletes, less water is available for seepage, and $\dot{m}_w$ decreases.
- The transition from saturated to partially saturated soil changes the effective evaporation area.

A more complete model would couple the mass transfer equation to the heat transfer:

$$\dot{m}_w(t) = h_m\, A_{\text{outer}} \left[\rho_{v,\text{sat}}(T_4) - \rho_{v,\infty}\right] \tag{T3}$$

T4. No thermal energy storage in the walls. Because of the quasi-steady assumption (T1), the walls have zero effective thermal capacitance in the model.

The only thermal mass is the milk:

$$C_{\text{total}} \approx C_{\text{milk}} = m_{\text{milk}} \cdot c_{p,\text{milk}} \tag{T4}$$

In a more detailed transient model, the wall thermal mass $C_{\text{wall}} = m_{\text{wall}} \cdot c_{p,\text{wall}}$ would appear, creating additional time constants. For clay walls of comparable mass to the milk, this could slow the overall thermal response by 20–40%.

T5. No phase change in milk. The milk remains liquid throughout the process (no freezing, no boiling). Given the temperature range (20–45 °C), this is trivially satisfied.

14.6 Evaporative Cooling Assumptions

E1. Evaporation occurs only at the outer surface. Water seeps through the porous outer soil wall and evaporates from the external surface $r = r_4$. We assume no evaporation occurs from the inner soil wall into the milk cavity or from the top/bottom surfaces.

E2. Evaporative heat is drawn entirely from the outer surface. The latent heat $\dot{m}_w h_{fg}$ is extracted from the outer surface of the container, not from the ambient air. This is the standard assumption for wet-surface evaporative cooling and is justified because the phase change occurs at the surface–air interface, cooling the surface.

E3. Humidity factor as a simple linear scaling. The effect of ambient relative humidity ϕ on evaporation rate is modeled as:

$$\dot{m}_{w,\text{effective}} = \dot{m}_w \cdot (1 - \phi) \tag{E3}$$

This is a first-order approximation. The rigorous expression involves the difference between the saturation vapor pressure at the surface temperature and the partial pressure of water vapor in the ambient:

$$\dot{m}_w = h_m\, A\, \frac{M_w}{R_u T} \left[p_{\text{sat}}(T_4) - \phi\, p_{\text{sat}}(T_\infty)\right] \tag{E3b}$$

The linear approximation is reasonable when $T_4 \approx T_\infty$ and ϕ is moderate.

E4. Sufficient water supply throughout the process. We assume the straw+water reservoir contains enough water to sustain evaporation for the full 7 hours. If $\dot{m}_w = 200\,\text{g/hr}$, total water consumed is $\approx 1.4\,\text{kg}$. The design must ensure the annular gap holds at least this much water initially.

E5. Uniform seepage through the wall. Water seeps uniformly through all parts of the outer soil wall, resulting in uniform evaporation over the entire outer surface. In practice, gravitational effects cause more water to accumulate at the bottom, creating non-uniform seepage and preferential evaporation from the upper (drier) regions.

E6. No mass transfer resistance through the soil wall. The rate of water seepage through the porous soil wall to the outer surface is assumed to be fast enough to sustain the evaporation rate. If the soil permeability is too low, the surface dries out and evaporation is limited by the seepage rate rather than the atmospheric demand.

14.7 Milk and Biological Assumptions

M1. Milk is a single-phase, homogeneous Newtonian fluid. Milk is actually a colloidal suspension (fat globules, casein micelles, whey proteins in water). We treat it as a homogeneous liquid with bulk properties (ρ, c_p, k). This is standard in food engineering and introduces negligible error for heat transfer.

M2. No heat generation from microbial activity. Bacterial metabolism in milk produces a small amount of heat. At the temperatures considered ($< 40\,^\circ$C) and the timescale (7 hours), this metabolic heat is negligible compared to the heat fluxes from conduction and evaporation (~ 0.01–0.1 W vs. ~ 10–100 W through the walls).

M3. Milk mass remains constant. No milk is removed during the 7-hour period (no deliveries reduce the volume partway through). In the real scenario, the milkman delivers portions to customers, reducing $m(t)$. Each delivery would be a step change in thermal mass $m \cdot c_p$, causing the time constant τ to decrease and the remaining milk to respond faster to the ambient.

M4. Initial milk temperature $T_{m,0} = 37\,^\circ$C. Fresh cow's milk exits at the cow's body temperature $\approx 38.5\,^\circ$C. We assume ≈ 1–$2\,^\circ$C of cooling during the brief collection and pouring process, giving $T_{m,0} \approx 37\,^\circ$C.

M5. Milk does not interact chemically with the soil wall. No dissolution, absorption, or reaction between the milk and the inner clay surface is considered. If the clay is not food-safe or is unglazed, minerals could leach into the milk, but this does not affect the thermal model.

14.8 Summary: Hierarchy of Assumptions

The assumptions are listed below in order of their *impact* on accuracy. Those at the top, if violated, would produce the largest deviation from reality:

Table 5: Assumptions ranked by impact on model accuracy

Rank	ID	Assumption	Impact
1	T3	Constant evaporation rate $\dot{m}_w$	High
2	B3	No direct solar radiation	High
3	T2	Lumped milk (Bi $\not\ll 1$)	Moderate–High
4	P1	Constant k_{soil} (ignores moisture variation)	Moderate
5	H5	Constant h_{conv}, h_{rad}, h_{milk}	Moderate
6	B4	Adiabatic top/bottom (no axial heat transfer)	Moderate
7	T4	No wall thermal capacitance	Moderate
8	E5	Uniform seepage and evaporation	Low–Moderate
9	H6	No forced convection during transport	Low–Moderate
10	B1	Linear ambient temperature rise	Low
11	G1	1-D radial (infinite cylinder)	Low
12	P5	h_{fg} at 100 °C instead of ~ 25 °C	Low
13	M2	No metabolic heat from bacteria	Negligible
14	P2	Constant milk properties	Negligible
15	M5	No milk–clay interaction	Negligible

15 Future Prospects for High-Strength Earthen Materials

The present soil-based evaporative milk carrier successfully demonstrates passive evaporative cooling, low thermal conductivity, eco-friendly operation, biodegradable material usage, and sustainable thermal insulation.

However, the current earthen structure still suffers from several engineering limitations including brittleness, heavy weight, low mobility, water sensitivity, and low tensile strength.

The future goal of this project is therefore to develop **next-generation engineered earthen composites** that preserve:

- soil breathability,
- evaporative cooling capability,
- thermal comfort,
- environmental sustainability,
- and natural porosity,

while simultaneously achieving:

- high strength,
- lightweight behavior,
- enhanced durability,
- and high mobility.

15.1 Engineering Limitations

Drawback	**Engineering Issue**
Brittleness	Sudden cracking under impact or stress
Heavy weight	Difficult transportation and handling
Low mobility	Reduced portability for rural transport
Water sensitivity	Structural weakening after prolonged wetting
Low tensile strength	Poor resistance to bending and shock
Fragility during transport	Risk of fracture under vibration

15.2 Fiber-Reinforced Earthen Composites

Natural fibers can act as reinforcement inside the soil matrix, similar to reinforcement in concrete.

Possible Reinforcement Materials

Natural Fiber	Expected Benefit
Jute fiber	Crack arrest and toughness
Coir fiber	Flexibility and impact resistance
Hemp fiber	Tensile strengthening
Bamboo fibers	Structural rigidity
Straw reinforcement	Lightweight porosity
Banana fibers	Sustainable reinforcement

Expected improvements include:

- reduced brittleness,
- higher tensile strength,
- improved impact resistance,
- lower crack propagation,
- and improved vibration tolerance.

15.3 Lightweight Earthen Materials

Future designs should aim to reduce the overall container weight while preserving thermal insulation and evaporative cooling capability.

Lightweight Additives

Additive	Function
Rice husk ash	Reduces density
Sawdust	Creates porous lightweight structure
Biochar	Thermal insulation and low density
Perlite	Lightweight thermal filler
Vermiculite	Heat insulation enhancement
Expanded clay	Lightweight aggregate

Engineering advantages include:

- easier transportation,
- improved portability,
- reduced handling effort,
- lower structural loading,
- and better rural mobility.

15.4 Eco-Binder and Geopolymer Stabilization

Pure soil possesses limited compressive strength and poor water resistance. Future systems may therefore incorporate eco-friendly stabilizers and geopolymer binders.

Possible Eco-Binders

Stabilizer	Role
Lime	Soil stabilization
Fly ash	Pozzolanic strengthening
Geopolymers	High compressive strength
Natural pozzolans	Durability improvement
Magnesium binders	Low-carbon alternative

Expected improvements include:

- higher compressive strength,
- improved moisture resistance,
- increased durability,
- and longer operational life.

15.5 Bamboo and Natural Mesh Reinforcement

Internal reinforcement meshes may improve the load-bearing capability of earthen structures.
Potential reinforcement systems include:

- bamboo lattice structures,
- jute mesh layers,
- hemp reinforcement sheets,
- natural rope frameworks,
- and biopolymer reinforcement grids.

Benefits include:

- prevention of catastrophic brittle failure,
- enhanced structural integrity,
- improved vibration resistance,
- and safer transportation.

15.6 Hybrid Soil–Ceramic Composites

Partially fired or semi-ceramic earthen materials may improve durability while retaining evaporative cooling capability.

Property	Improvement
Hardness	Increased
Crack resistance	Improved
Water resistance	Improved
Thermal insulation	Maintained
Cooling capability	Preserved

15.7 Nano-Engineered Earthen Materials

Nano-scale additives may significantly strengthen particle bonding inside soil composites.

Nanomaterial	Function
Nano-clay	Improved particle packing
Cellulose nanofibers	Natural reinforcement
Silica nanoparticles	Crack reduction
Graphene oxide	Mechanical strengthening

15.8 High-Mobility Modular Designs

The present design has limited portability due to high weight and brittle walls.

Future Mobility Improvements

Proposed Modification	Purpose
Detachable wall sections	Easier repair
Modular assembly	Better transport
Wheel-supported base	Improved movement
Composite carrying frame	Reduced wall stress
Lightweight layered shells	Lower overall mass

15.9 Self-Healing Earthen Materials

Bio-mineralizing microorganisms may automatically seal cracks inside earthen materials through mineral precipitation.

Advantages include:

- increased service life,
- reduced maintenance,
- improved durability,
- and higher reliability.

15.10 Water-Resistant Breathable Coatings

The material must remain breathable for evaporation while resisting structural erosion.

Coating	Purpose
Natural wax	Water resistance
Linseed oil	Surface protection
Clay–lime coating	Breathable barrier
Bio-hydrophobic coating	Moisture protection

15.11 Future Applications

Potential applications include:

- sustainable housing materials,
- passive cooling walls,
- rural refrigeration chambers,
- eco-friendly food storage,
- low-carbon construction panels,
- thermal insulation systems,
- disaster-relief shelters,
- and sustainable urban housing.

15.12 Long-Term Research Vision

The ultimate objective is to create:

Smart Sustainable Earthen Engineering Materials

that combine:

Desired Property	Target
High strength	Comparable to concrete
Low density	Lightweight structure
Thermal insulation	Passive cooling
Environmental sustainability	Low carbon footprint
Breathability	Natural humidity control
Durability	Long service life
Mobility	Easy transportation
Eco-friendliness	Biodegradable and non-toxic

15.13 Future Work

- Develop experimentally validated lightweight earthen composites.
- Measure compressive, tensile, and flexural strength.
- Optimize porosity for cooling and structural integrity.
- Study long-term moisture durability.
- Develop hybrid soil–fiber–geopolymer composites.
- Design modular high-mobility container systems.
- Investigate self-healing bio-earthen materials.
- Explore large-scale sustainable construction applications.

References

1. M. Cappai, A. Nait-Ali, and S. Sfarra, “Thermal Properties of Eco-Friendly Earthen Materials Stabilized with Bio-Based Polymers,” *Materials*, vol. 17, no. 5, pp. 1035–1052, 2024.

2. M. Syed, M. Ashfaq, B. Jamhiri, U. Ali, and F. E. Jalal, “Sustainable Engineering of Fiber-Reinforced Geopolymer-Treated Low Plasticity Clay Linking Geomechanics and Microstructure Through Support Vector Machines,” *Scientific Reports*, vol. 15, no. 1, pp. 44399, 2025.

3. F. Ben, H. Rahmouni, and M. Saidi, “Investigation of the Thermal Properties of Clay Composites Reinforced with Biomass and Polymer Waste,” *Results in Engineering*, vol. 25, pp. 102010, 2026.

4. W. Fouad, M. Marvila, and A. Azevedo, “Thermal Insulation and Mechanical Performance of Stabilized Rammed Earth Incorporating Sustainable Additives,” *Scientific Reports*, vol. 15, pp. 30472, 2025.

5. B. Aksu Alcan, E. Demir, and M. Yilmaz, “Geopolymerization of Kaolin Clay with Hemp Fibers for Sustainable Composite Materials,” *Polymers*, vol. 17, no. 23, pp. 3216–3235, 2025.

6. Y.-J. Du, B.-W. Yu, K. Liu, and N.-J. Jiang, “Physical, Hydraulic, and Mechanical Properties of Clayey Soil Stabilized by Lightweight Alkali-Activated Slag Geopolymer,” *Journal of Materials in Civil Engineering*, vol. 32, no. 6, pp. 04020115, 2020.

7. J. Davidovits, *Geopolymer Chemistry and Applications*, 5th ed., Saint-Quentin, France: Institut Géopolymère, 2020.

8. G. S. Ryu, Y. B. Lee, K. T. Koh, and Y. S. Chung, “The Mechanical Properties of Fly Ash-Based Geopolymer Concrete with Alkaline Activators,” *Construction and Building Materials*, vol. 47, pp. 409–418, 2013.

9. G. Mihalakakou, M. Souliotis, M. Papadaki, and G. Halkos, “Applications of Earth-to-Air Heat Exchangers: A Holistic Review,” *Renewable and Sustainable Energy Reviews*, vol. 152, pp. 111715, 2021.

10. F. Gouny, F. Fouchal, P. Maillard, and S. Rossignol, “Study of the Effect of Siliceous Species in the Formation of a Geopolymer Binder: Understanding the Reaction Mechanisms Among the Binder, Wood, and Earth Brick,” *Industrial & Engineering Chemistry Research*, vol. 53, no. 9, pp. 3556–3565, 2014.

11. Y. A. Çengel and A. J. Ghajar, *Heat and Mass Transfer: Fundamentals and Applications*, 6th ed., New York, NY, USA: McGraw-Hill Education, 2020.

12. J. P. Holman, *Heat Transfer*, 10th ed., New York, NY, USA: McGraw-Hill Education, 2010.

Appendix

Practical & Commercial Questions

Q1. What if milk reacts with the soil and microbacteria develop, causing spoilage?

Microbial contamination is an important food-safety concern for any porous earthen storage system. Since milk naturally contains bacteria that grow rapidly between $4°C$ and $60°C$, prolonged exposure to warm conditions may increase spoilage rates. In addition, untreated soil surfaces may absorb moisture and create favorable conditions for bacterial growth.

To overcome this issue, the proposed design incorporates a **food-safe inner protective coating** while maintaining the evaporative cooling capability of the outer wall. The inner surface can be coated with:

- Food-grade antibacterial ceramic coating,
- Natural beeswax lining,
- Silver-ion based antimicrobial coating,
- Lime-based antibacterial paint.

These coatings prevent direct milk-soil interaction, reduce bacterial adhesion, and minimize contamination risks. The outer surface remains porous to allow evaporation and cooling. In addition, periodic cleaning, sterilized water usage, and daily drying of the straw layer further improve hygiene.

Hence, the container is designed not as a permanent preservation system, but as a **safe short-duration rural transport solution** that delays spoilage while remaining environmentally sustainable.

Q2. How is this product cost-effective, and why would milkmen shift from aluminium containers to the Soil-Based Evaporative Milk Carrier?

Conventional aluminium milk containers provide almost no thermal insulation. During summer transport, milk temperature rises rapidly, increasing spoilage risk and reducing milk quality. In contrast, the Soil-Based Evaporative Milk Carrier passively cools the milk using evaporation without requiring electricity or refrigeration.

The proposed carrier offers several advantages:

- Lower milk spoilage during rural transportation,
- Zero electricity consumption,
- Reduced refrigeration dependency,
- Eco-friendly and biodegradable materials,
- Low manufacturing and maintenance cost,
- Better milk freshness over 4–7 hour transport windows.

Although the initial cost may be slightly higher than a basic aluminium container, the reduction in spoiled milk and refrigeration expenses makes the system economically beneficial in the long term. Since the container is made primarily from locally available clay, straw, and water-retaining materials, rural production costs remain low.

Therefore, milkmen may prefer this system because it improves milk quality, reduces losses during transportation, and provides a sustainable alternative to metallic containers in regions with limited refrigeration access.

Q3. How portable is the product, and what is its mobility?

The Soil-Based Evaporative Milk Carrier is designed specifically for rural and semi-urban transportation conditions. The cylindrical geometry provides structural stability, while the compact design allows easy portability on bicycles, motorcycles, handcarts, and small transport vehicles commonly used by milk vendors.

To improve mobility, the design can include:

- Side handles for manual carrying,
- Lightweight reinforced outer frame,
- Protective shock-absorbing base,
- Lockable insulated lid,
- Modular sizes for 5 L, 10 L, and 20 L capacities.

Even though earthen materials are heavier than aluminium, the improved thermal performance compensates for the additional weight by preserving milk quality during transport. Furthermore, the container does not require external power supply or refrigeration equipment, making it highly practical for remote villages and long-distance rural delivery routes.

Hence, the product provides a balance between portability, thermal protection, durability, and rural usability.